

[CH5] The Computer Becomes a Business Machine

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- In the 1950s the computer was starting to transform into an electronic data-processing machine instead of a mathematical instrument
- In 1951, IBM recognized this and rapidly reoriented its resources. Within 5 years, it dominated the market.
- In the US about 30 companies entered the computer business. In the UK about 10, but they had few customers. The US dominated the market
- 3 types of companies were present:
 - Electronics and control equipment manufacturers:
 - Like RCA, General Electric, Raytheon, Philco, Honeywell, Bendix
 - Used to selling high-value electronics => The computer was "natural" to them
 - Coined the term **mainframe**
 - The computer was just another element in their high-value product lineup
 - Business machine manufactureres
 - Like IBM, Remington Rand, Burroughs, NCR, Underwood, Monroe, Royal
 - Lagged a few years behind the first group, because it wasn't as natural for them to enter the mathematical computing instrument business. After Eckert and Mauchly's UNIVAC in 1951, they entered the race
 - Entrepreneurial start-ups:
 - In the beginning (1948-1953) it was relatively cheap to join the computer industry (under 1 million USD). A decade later this cost would be 100 million
 - The first entrepreneurial firms were Eckert and Mauchly's Electronic Control Company (1946), and the Engineering Research Associates (1946) founded by William Norris and others.
 - Following them came a rush of start-ups: Electrodata, Computer Research Company, Librascope, Laboratory of Electronics
 - Some lucky ones were taken over by office-machine companies, others faded away

UNIVAC & BINAC

When Eckert and Mauchly set up the ECC there were few venture capital firms. To get funding for the development of their machines, they needed an order.

The Census Bureau agreed to buy a computer for 300,000 USD. Eckert and Mauchly thought the cost of development would be 400,000 USD. In fact, the UNIVAC cost closer to 1,000,000 USD to develop

The UNIVAC stored its data on magnetic tape to replace the punched cards. However, the only commercially available tape decks were analog, so they had to develop a special digital magnetic tape drive. They also had to develop tape that wouldn't stretch, tape coatings, machines that would transcribe punched cards to tape, etc.

To fund all of this, they would need more money, so more orders.

Northrop Aircraft Corporation needed a small airborne computer for the Snark long-range guided missile, so they ordered a new computer: the BINAC. A small data processor that didn't use magnetic tape, very different from the UNIVAC.

The American Totalisator, whose vice president was Henry Strauss, bought 40% of the shares in ECC, seeing the potential of the UNIVAC and offered further funding. With this, they did not have to worry about money as much anymore.

The BINAC was delivered in 1949 but was unreliable, as most early computers were. Northrop was

satisfied enough to pay the remaining 20,000 USD of the order.

They could now focus on the UNIVAC, for which they had 1,200,000 USD worth of orders. However, Henry Strauss died in a plane crash and American Totalisator lost interest in the EMCC. They requested their loans back and the EMCC was immediately thrown into disarray.

IBM: Evolution, not Revolution.

- By the mid 1950s, IBM was in a position of dominance
- In March 1945, IBM had funded the Watson Scientific Computation Laboratory at Columbia University, headed by Prof. Wallace Eckert (no relation to Presper Eckert) with the goal of building an electronic calculator better than the Harvard Mark I. This was the SSEC
- This was so that IBM would have a test bed for new ideas. It was never expected that the SSEC would be salable
- Meanwhile, IBM engineers were incorporating electronics into existing products to make them faster
- In 1947 they developed the CPC (card programmed calculator). It was fast, reliable and showed IBM that there was a market for scientific computing equipment. In the first half of the 1950s, it sold over 700 units.
- When the SSEC was finished in 1948 and displayed in the IBM headquarters in Manhattan, it was already obsolete. However, it attracted public awe and cemented IBM as a leader in the computer technology industry

MDC & TPM

- The MDC (Magnetic Drum Calculator) and TPM (Tape Processing Machine) were also under development in 1949
- The MDC used a large magnetic drum instead of mercury delay lines and would sell for a tenth of the price of a UNIVAC
- The TPM was more on par with the UNIVAC
- IBM didn't press on with either of these and they only reached market 5 years later

Defense Calculator

- With the start of the Korean War, Thomas Watson Jr. saw a market for large-scale scientific computers with defense contractors.
- This took away resources from the TPM, which allowed the UNIVAC to capture the market

The UNIVAC comes alive

- Eckert and Mauchly were looking for funding and tried to sell the company to IBM, but they refused
- Remington Rand bought EMCC and Eckert and Mauchly became employees within it.
- By the beginning of 1951 the UNIVAC was functioning but needed software. Grace Murray Hopper became the driving force behind advanced programming techniques for commercial computers and the world's foremost female computer professional. She led the programming team.
- The Census Bureau accepted their UNIVAC in March 1951.
- In the 1952 presidential election, the UNIVAC gave a great early prediction of the result after the initial votes were in.
- UNIVAC was starting to become synonymous with computer.

IBM's Big Push:

- Terrified by the UNIVAC success, Watson Jr. decided to start a major push to counter the UNIVAC
- Development started at full throttle for the Defense Calculator (Model 701), the TPM (Model 702) and the MDC

- The 702 quickly outmarketed the UNIVAC. It used Williams Tube technology instead of mercury delay lines. This was more reliable and twice as fast. Its tape technology was also far superior.
- IBM computers were also modular. This made shipment easier and gave flexibility in regard to the size and spec of the machine.
- By 1953, core memory was in development. This was to replace the Williams Tube memories. Updated models of the 701 and 702, named the 704 and 705 were announced in 1954
- 1955 the orders for 700 series computers exceeded those of UNIVAC's
- It was, however, not the 700 series but the low cost MDC (named the Model 650 and announced in 1953) that secured IBM's leadership in the industry. It was cheap compared to the 700 series but twice as expensive as competitors from other vendors. However, it succeeded because of superior engineering, reliability and software.